

Commodity Boom and Industrialization: Brazil before WWII

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Abstract

Under what conditions do commodity booms lead to industrialization? We study this question in the context of Brazil in the 19th and early 20th centuries, when the region of São Paulo experienced a coffee boom together with the formation of its first industrial cluster. We digitize a host of new data on sectoral employment; agricultural, textile, and manufacturing censuses; trade flows (internal and external), and railroad investments for 1872-1950. The data show that regions with higher coffee suitability experienced slower manufacturing growth, while regions in proximity to those where coffee was booming experienced substantial manufacturing growth. We develop a quantitative model of Dutch Disease with a Big Push mechanism and show that the coffee boom can lead to industrialization, but only in the presence of substantial migration and low domestic trade costs.

Keywords: International trade, industrialization, agriculture, commodities, structural change

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1 Introduction

What is the impact of a commodity boom on industrialization? A common policy concern is that during a boom, countries will divert their productive factors away from industrial activity to produce commodities—a phenomenon often referred to as “Dutch disease”. Such concerns have been especially prominent in the study of the 19th and 20th centuries, when industrialization in Western Europe was associated with rapid economic growth, while the rest of the world specialized in the production of commodities (see, e.g., Williamson, 2011). Yet, during this same period, several industrial centers emerged against the backdrop of booming commodity sectors, suggesting that under the right conditions such booms can foster rather than prevent industrialization. Whether commodity booms crowd out or catalyze industry remains an open question in contemporary policy debates, where the risk of Dutch disease continues to shape discussions over trade and industrial policy across both developing and advanced economies.

In this paper, we study the conditions under which commodity booms can lead to industrialization. The empirical setting is the early development of Brazil’s manufacturing sector, from the end of the 19th century to World War II. During this period, São Paulo’s industrial sector grew at breakneck speed and overtook Rio de Janeiro—the capital city and hitherto main manufacturing center—and became what is nowadays one of the largest industrial clusters in South America. The inception of São Paulo’s manufacturing sector came on the heels of a boom in global coffee demand, during which Brazil supplied half of world production. Due to its ideal geographic conditions, São Paulo’s hinterland alone supplied one third of world production. We argue that the conditions created by coffee—demand growth, inflows of migrants, and the development of a domestic base—allowed for manufacturing to take off.

We begin by putting together a wide array of newly digitized data on Brazil between 1872 and 1950. Using population, agricultural, textile, and manufacturing censuses, together with domestic and international trade records, and the development of the railroad network we build a data set that allows us to characterize the birth of industry in the southeast of Brazil, as well as the relation between industry and coffee. We organize our empirical analysis into a set of macro facts and a set of micro facts, which we describe now.

We establish the following macro facts: (i) Concomitant with its growing presence in world coffee markets, the share of manufacturing in São Paulo’s exports to the rest of Brazil grew from virtually zero to 80 percent, while the share in imports plummeted; (ii) São Paulo’s Revealed Comparative Advantage in Manufacturing (relative to the rest of Brazil) grew as well, reflecting changes in relative costs, (iii) São Paulo experienced faster structural change than the rest of Brazil, with sharp movements of employment out of agriculture and into

manufacturing and services; (iv) São Paulo’s population grew faster than the rest of Brazil, bolstered by immigration of foreigners, and (v) Brazil’s imports of textiles dry up, replaced by machinery. Our macro facts point to a transformation of the São Paulo region in which domestic markets played a key role as destination of São Paulo’s manufacturing products.

Turning to micro data at the municipality level, we document a dual, causal role for coffee production suitability, as measured by the availability of terra roxa, a type of soil where coffee grows easily and which was one of the main drivers of São Paulo’s role in the coffee boom. We define two variables: “Own share” of terra roxa and “market access” terra roxa, i.e., access to regions with terra roxa, inversely weighted by trade costs (in the spirit of Harris, 1954). We then interact each of these variables with Brazil’s aggregate exports of coffee—instrumenting when possible with coffee production of other countries in the world. Our results suggest, on the one hand, that a region’s own coffee suitability (“own share”) is associated with specialization of local employment in agriculture, faster growth of foreign population and total population, slower development of the textile industry (numbers of firms, workers, looms); it is also associated with lower manufacturing productivity by the end of the period. On the other hand, “market access” to terra roxa was associated with faster manufacturing employment growth, faster foreign immigration (but no faster population growth), faster development of the textile industry, and higher manufacturing productivity by the end of the period. We interpret these findings as evidence that proximity to other regions benefited by the boom created a domestic demand base for manufacturing production, while coffee suitability itself fostered the coffee industry.

We next develop a quantitative model of domestic and international trade and migration. Our quantification has two goals. The first is to understand whether, in a model disciplined by our micro facts, a coffee boom can generate a transformation of the magnitude documented in our macro facts. This is the premise of a large historical literature in Brazil, which has tied coffee to industry to various channels (Dean, 1969; Summerhill, 2005; Suzigan, 1986). The second goal is to demonstrate that to match the data, a model must feature microfoundations for manufacturing productivity growth that deviate from standard neoclassical assumptions as well as standard formulations of increasing returns to scale.

Our model features many heterogeneous regions (including a foreign economy) and four sectors: coffee, food, manufacturing, and services. It allows for domestic and international trade and migration. We calibrate our model to the year 1920 and evaluate a series of counterfactual scenarios. Across all scenarios, we consider the effect of a drop in coffee demand from the rest of the world, so as to bring Brazil’s exports to their level in 1872. We consider four variants: (i) only the drop in world coffee demand occurs, and then in addition (ii) there are Marshallian externalities in the manufacturing sector, (iii) there is a

simultaneous, exogenous drop in São Paulo’s manufacturing productivity, and (iv) there is a simultaneous drop in manufacturing productivity, correlated with market access to terra roxa (as in our micro facts). Our results show that, absent a simultaneous shock in productivity, a purely neoclassical model is unable to relate the coffee boom to manufacturing development (even in the presence of Marshallian externalities). That is, such a model cannot rationalize all our micro and macro facts without an additional manufacturing productivity shock.

In the last section of the paper, therefore, we turn to a simplified version of our model in which there is the potential for Big Push adjustment. The model features two main changes relative to our quantitative model above: it allows for the kind of industrialization introduced by Murphy, Shleifer, and Vishny (1989b) and it simplifies to three regions: São Paulo, Rest of Brazil and Rest of the World. The model drives home three key messages. First, the coffee boom can eliminate a bad, no-industrialization equilibrium by generating local demand for manufacturing goods. Second, migration is essential to allow for both the coffee and manufacturing boom to occur, as it relaxes factor endowment constraints. Third, internal trade played an important role, by allowing for manufacturing to concentrate in São Paulo and serve the whole country as its industrial base.

Our paper contributes to several strands of literature. First, we contribute to the debate on how commodity booms relate to industrial development. In seminal work, Corden and Neary (1982) formalized the notion of Dutch disease, while Matsuyama (1992) showed how comparative advantage might lock countries into slow-growth sectors. In this work, booms lead to stagnant manufacturing, or even deindustrialization. Other work has studied the possibility that commodity booms might generate scale for industrialization, a possibility studied theoretically by Murphy, Shleifer, and Vishny (1989a), and highlighted in earlier work in economic development, including Rosenstein-Rodan (1943), Nurkse (1952), Lewis et al. (1954), and Ranis and Fei (1961). Recently, Buera, Hopenhayn, Shin, and Trachter (2026) have studied the interaction of Big Push with misallocation. Our contribution to this debate is to document the role of domestic markets and trade integration, both through infrastructure and demand, in propelling an industrial sector.

Our paper contributes to the literature discussing the role of agriculture in the process of development, such as Johnston and Mellor (1961), Gollin, Parente, and Rogerson (2002), and Gollin, Jedwab, and Vollrath (2016). Different from Gollin, Jedwab, and Vollrath (2016), who argue that African countries subject to commodity booms urbanize without industrialization, here we investigate a case in which the commodity boom generated the necessary conditions for the formation of a local industrial sector. More recently, Bustos, Caprettini, and Ponticelli (2016) and Bustos, Garber, and Ponticelli (2020) have documented agricultural productivity growth relates to manufacturing through labor markets and investment. Our results suggest

that the commodity boom created a domestic manufacturing base that enabled local and aggregate industrialization.

A more recent literature studies the interactions between structural change and economic geography. This literature includes, among others, Fajgelbaum and Redding (2022), Uribe-Castro (2021), Clark, O’Rourke, and Taylor (2008), Fan, Peters, and Zilibotti (2021), and Eckert and Peters (2022). The key distinguishing feature of our paper is to show how external shocks can interact with a domestic home market effect to generate industrialization. The paper also relates to the literature on endogenous infrastructure investments (Fajgelbaum and Schaal, 2020 and Santamaria, 2020). Different from these papers, we study the cross-sectoral spillovers that arise when infrastructure is optimally invested to support one sector.

Lastly, this paper contributes to growing research on trade and economic history, applying spatial equilibrium models. Computational equilibrium models have a long tradition in economic history — see, e.g., O’Rourke and Williamson (2001) for several applications related to the globalization period in the end of the 19th century. Recent quantitative spatial equilibrium models include Nagy et al. (2016) and Pellegrina and Sotelo (2025).

2 Coffee Boom and Industrialization in Brazil pre-WWII

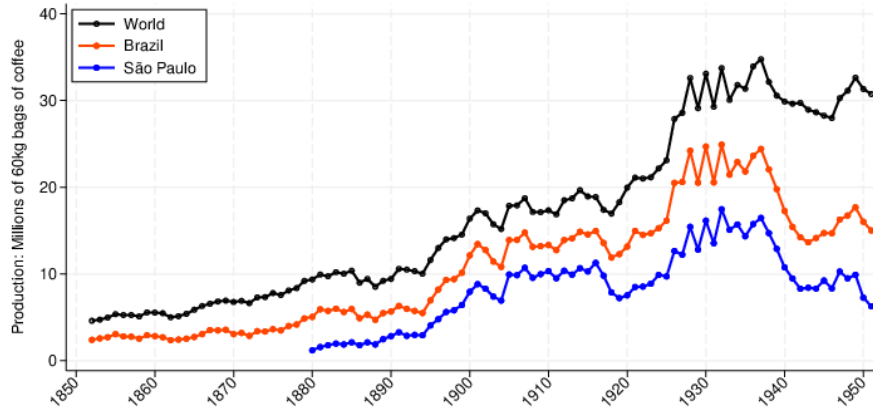
This section describes the economic and institutional context of Brazil’s coffee expansion before World War II, focusing on the transformation of the state of São Paulo, from a sparsely populated rural area into the country’s first major industrial cluster. We start by describing the coffee expansion and then present the figures related to the industrialization. In the next section, we present new facts about the coffee expansion using the newly digitized data that we collected.

2.1 Coffee Expansion, Globally and in Brazil

Before the development of a large domestic industry in the 20th century, Brazil’s trade relationships with the rest of the world was marked by large cycles of commodity booms. Its first cycle started with a rise of export of an abundant wood called “Pau Brazil,” which gave the country its name, it then moved sequentially to sugarcane between the 16th and the 18th century, gold in the late 17th until the 19th century, and finally to coffee in the middle of the 19th century.¹

¹In the beginning of the 20th century, Brazil also experienced a large, but temporary, rubber cycle concentrated in the Amazon forest.

Figure 1: Total Coffee Production in the World, Brazil and in the State of São Paulo



The first coffee plantation in Brazil was established in the far north region of the country, in the Amazon forest, with plants brought from French Guiana (Ukers, 1922).² However, it was not until the middle of the 19th century that coffee became a major exporting commodity in Brazil, with most of the production coming from plantations located in the State of Rio de Janeiro in the southeast of the country. Over time, coffee plantations moved southwest from Rio de Janeiro, skirting the steep mountains of the State of Minas Gerais, to the region called Vale do Paraíba in the State of São Paulo, where the Paraíba river could be used to ship out the coffee through the port of Rio de Janeiro. While coffee farms were present in São Paulo as early as 1798, it was not until the end of the 1890s when coffee plantations would reach the São Paulo’s western plateau that would later become the heart of the coffee production in Brazil, including municipalities such as São Carlos, Campinas, and Rio Claro (Luna and Klein, 2003).

Figure 1 illustrates the expansion of coffee production during this period, in Brazil and the rest of the world. Between 1870 and 1920, Brazil’s coffee exports increased five fold and its share of world production reached 80 percent. The bulk of this expansion came from the State of São Paulo, which reached almost 60% of Brazil’s total by 1920. This expansion deeply transformed São Paulo’s political, cultural and economic landscape. Once a peripheral state, São Paulo emerged as a leading political force, alternating the presidency with Minas Gerais under the arrangement known as “Café com Leite” (“coffee with milk”). Coffee wealth also financed major cultural initiatives, culminating in the *Semana de Arte Moderna* (“week of modern art”), from which members of elite coffee-growing families went on to become internationally renowned artists—for example, the painter Tarsila do Amaral.

²Before the expansion of coffee in Brazil, the largest global producer was the french colony of Saint-Domingue, in the area of modern Haiti. By 1780s, Saint-Domingue produced 60 percent of all coffee consumed in Europe.

Economically, by 1920 coffee exports accounted for roughly half of Brazil’s total exports and as much as 87 percent of São Paulo’s exports.

To put these numbers in perspective, the global population roughly doubled during this period, whereas global coffee production increased sixfold. Much of this expansion in consumption reflected changing habits and a growing appreciation for coffee in destination markets in Europe and the United States, for example related to the Belle Époque.

2.2 Geographic and Economic Conditions for Coffee’s Takeoff

Three main conditions enabled the takeoff of coffee in Brazil and the subsequent industrialization of São Paulo: The concentrated availability of highly suitable soil, called Red Latosoil or “Terra Roxa”; the expansion of the railroad network, radiating out of São Paulo’s port of Santos, and a large inflow of foreign migrants, primarily into the São Paulo area. We explain each in turn.

Red latosolos. One of the reasons for the successful expansion of coffee in the State of São Paulo is its agro-climatic conditions. Located just above the tropic of Capricorn, the state is mostly dominated by a tropical climate that avoids morning frosts, which are highly damaging to coffee trees—weather conditions in low latitude regions still keep coffee plantations from expanding down to the south region of Brazil to this day. Another key agro-climatic condition of the State is its abundance in a type of soil called red latosolos, which is also commonly known in Brazil as the “purple soil”, a highly suitable type of soil for the production of coffee due to its high content of iron and aluminium oxides.³ As we document later in Section 4, the geographic expansion of coffee is indeed strongly correlated with the presence of such type of soil.⁴

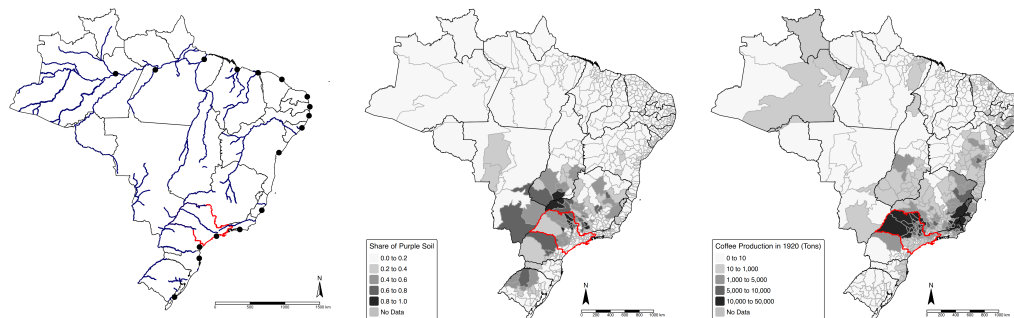
Transportation costs. Until the arrival of the railroad interior region of the State of São Paulo were particularly poor. The Atlantic Coast and the interior plains of the State where coffee plantations could be established are cut by a high and steep sea mountain area called

³The existence of two names for this type of soil, *red soil* and *purple soil*, was caused by a false cognate between Italian and Portuguese. When Italians arrived to produce coffee in Brazil, they would call the soil *terra rossa*, which in Italian means *red soil*, due to its strong red coloration (see Appendix Figure 12). In Portuguese, *terra rossa* is closer phonetically to *terra roxa*, which in Portuguese means “purple soil”, instead of *terra vermelha*, which actually means “red soil” in Portuguese. In Brazilian old soil classification, the official name of this soil was *purple soil*, but recently it has changed into *red latosolo*.

⁴In addition to the nutritious soil for coffee, historians also mention the large availability of land, which allowed farmers to space out coffee trees more widely than in other parts of the world, which allowed for higher labor productivity. For example, according to Marquese (2020), an enslaved worker in Brazil would cultivate twice as many coffee bushes than a worker in the Caribbean area.

Figure 2: Coffee: Economics and Geography

(a) Navigable Rivers and Ports (b) Terra Roxa availability (c) Coffee Production in 1920



“Serra do Mar”, with no river flowing from the interior plains to the coast. As documented by Summerhill (2005), due to this geography, transportation in the State of São Paulo in the pre-rail era often went on the back of mules—by 1914, carrying charges for transportation via mules exceeded that of railroads by an order of magnitude.⁵ Indeed, nearly 20% of a coffee plantation’s workforce (often composed of enslaved people) could be committed to transportation related services at any given time (Hall, 1969). As such, before the arrival of railroads, it was prohibitively expensive to ship coffee at large scales directly from São Paulo to the rest of the world.

The expansion of coffee production into São Paulo would coincide with the expansion of its railroad stock. As early as 1850 it was understood that São Paulo’s rich red soils were unusually well suited for producing coffee. This is demonstrated by the success of Areias, a São Paulo county near the Paraíba Valley, at growing coffee prior to the rest of the state (Luna and Klein, 2003). The construction of the state’s railroads began in the late 1860s, but it was the following decade that saw the railway stock explode. About 2600 *km* of railway was built in Brazil by the end of the 1870s and almost all of this was in São Paulo’s coffee growing region (Hall, 1969). Soon after this initial explosion in rail lines, in 1883, when around 6000 *km* of railroads had been laid, São Paulo would match Rio’s coffee production for the first time (Leff, 1982; Stein, 1957). This expansion in rail infrastructure would continue into following decades. Appendix Figure 11 shows the large increase in the total length of the railroad network until 1910 in the state. This expansion connected the interior regions of the State of São Paulo to the port of Santos, from which the coffee could be shipped out to the rest of the world.

⁵The government had a central role in the expansion of the railroad. In addition to directing the location of the railroads through concessions, the government itself owned, according to Summerhill (2005), 61% of the railroad system by 1914.

Immigration. To sustain the expansion of coffee plantations in São Paulo, there was a massive wave of immigrants to the State, coming particularly from Italy (see Appendix Figure 13 and Figure 5 below). During the peak of this movement *circa* 1920, approximately 18% of São Paulo’s population was foreign born. The immigration was facilitated by a series of public-private partnerships between the state of São Paulo and the coffee industry that began in 1884 and took off in 1886. The government of São Paulo subsidized the travel and housing of newly arrived immigrants and non-profits, funded by coffee growing fazendeiros, waged propaganda campaigns in Europe to encourage farmers to take advantage of these opportunities (see Holloway (2017) and Hall (1969) for a detailed discussion of these policies).⁶ A major motivation of these policies was to provide cheap labor after the abolition of Slavery in 1888 (Leff, 1972; Hall, 1969). By 1922, when this practice ended, São Paulo had encouraged between 750,000-2.8 million Europeans to travel to Brazil (Leff, 1982; Holloway, 1975).

3 Data

We collect and digitize data from several different sources to construct a panel dataset with information on labor employment, immigration, trade (internal and external), and access to railroads. Our final dataset includes 623 municipalities distributed across 22 states.⁷ Next, we describe the sources of data that we used.

First, we digitized data from all Brazil’s demographic census between 1872 and 1950, which includes the editions of 1872, 1890, 1900, 1920, 1940, and 1950. These collect information on the number of workers by sectors and nationality—whether the worker was born in Brazil—with the data being disaggregated at the municipality level for 1872, 1920, 1940 and 1950. We categorized the sectors into manufacturing, services, and agriculture.

Second, we bring in data from Brazil’s manufacturing censuses, which starts in 1907. We have information at the municipality level on the number of firms, total employment, capital stock, horsepower, and production value. At the state-level, we also have the same

⁶After the abolition of slavery in Brazil in 1888 with the “Lei Aurea”, governments searched for alternative forms of attracting labor to regions in economic expansion. Producers attempted to implement a sharecropping contract with immigrants, but that type of contract failed to attract immigrants. Following sharecropping, one of the main arrangements used in the region to attract immigrants was called “colonato”. In this arrangement, a “colono” family would contract with a “fazendeiro” (farm owner) to tend a fixed number of coffee trees and harvest them. These families could cultivate also food crops, which they could keep for themselves for consumption. It was not a pure wage nor sharecropping type of contract. It was a hybrid system that shifted some of the risk of production to the colono.

⁷Between 1872 and 1950 there were around 1200 new municipalities in Brazil. To ensure geographic units that are consistent over time, we aggregate municipalities into broader units according to the number of municipalities in 1872.

information disaggregated for each industry sector. We also digitize data from Brazil’s textile censuses for the years 1853, 1866, 1885, 1905, 1921, and 1945.

Third, we construct historical railroad maps for each decade between 1860 and 1950. We digitize data from multiple sources, including official government reports on the extent of the railroads and political and topographic maps of the Brazilian territory that contained the railroad network. In addition, we digitized historical accounting books with information on internal trade in different segments of the railroad system between 1901 and 1950.

Lastly, we also collected data on internal and external trade from Brazil’s official international trade reports between 1841 and 1950. We have information on volume and value of imported and exported goods at the state level. We also observe internal trade via coastal shipping, which was one of the main modes of transportation between Brazilian states.

4 Empirical Observations

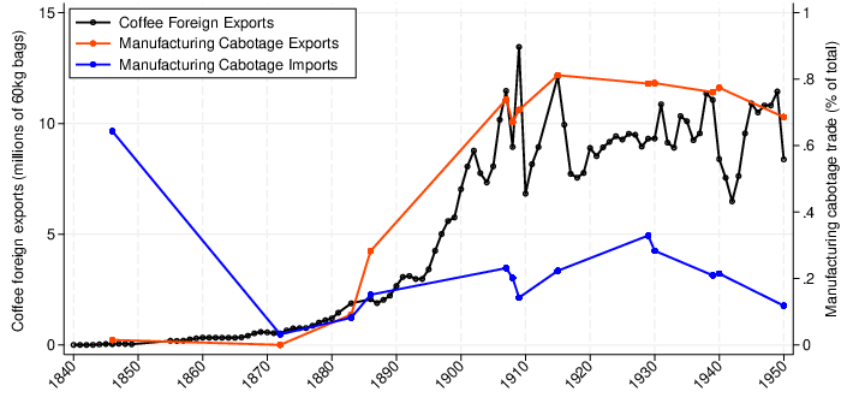
We now present two sets of empirical patterns that motivate the mechanisms we introduce in our spatial equilibrium model in Section 5. First, our macro facts document the large transformation that the São Paulo region experienced regarding domestic and international trade, migration, its specialization in manufacturing and its importance in the Brazilian economy. Second, our micro facts show that the coffee boom had two distinct effects across regions in Brazil, inducing specialization in agriculture in regions with high coffee suitability but inducing specialization in manufacturing in regions that were close to those that had high suitability in coffee.

4.1 Macro Facts

4.1.1 Coffee Boom and São Paulo’s Exports to the rest of Brazil

Worldwide growth in coffee consumption was a factor of 6 between 1879 and 1930. During this period, Brazil accounts for about 85 percent of production, and São Paulo for 50 percent. At the same time, there are large changes in São Paulo’s export and import basket from the rest of Brazil. The share of manufacturing imports drops sharply from nearly 80 percent to about 10 percent, while the share of manufacturing exports rose from virtually zero to about 80 percent.

Figure 3: Coffee Exports and Manufacturing Exports to the Rest of Brazil



4.1.2 São Paulo gained a comparative advantage in light manufacturing goods relative to the rest of Brazil

We begin by documenting a central fact in the evolution of the Brazilian economy. During the coffee boom, São Paulo became more specialized in manufacturing, relative to the rest of the country and relative to other regions of Brazil. Table 1 shows the evolution of São Paulo's Revealed Comparative Advantage between 1846 and 1940, for five broad industries, relative to the rest of Brazil. In particular, we compute

$$RCA_{s,k} = \frac{X_{s,k} / \sum_{k'} X_{s,k'}}{X_k^{BRA} / \sum_{k'} X_{k'}^{BRA}},$$

where $X_{s,k}$ denotes cabotage exports of state s in good k and X_k^{BRA} denotes the sum of cabotage exports in good k for all of Brazil.⁸ The table shows one key fact: During this period, São Paulo's exports to the rest of Brazil started specializing in manufacturing goods, while specialization in cash crops decreased sharply. Its specialization grew by a factor of 50 over this period. The table shows, for comparison, Rio de Janeiro's RCA. Bearing in mind that Rio de Janeiro was the capital city, with a more developed manufacturing industry, we note that its RCA grew only by a factor of 5.

4.1.3 Faster structural change and industrialization in São Paulo

This increasing specialization relative to the rest of Brazil had large implications for São Paulo's aggregate production structure. The trade flows are mirrored in São Paulo's employment, whose manufacturing share grew about twice as fast as the rest of the country, while

⁸Data unavailable for RJ in 1930 and 1940.

Table 1: São Paulo gained a Comparative advantage in Manufacturing between 1846 and 1940

	Sao Paulo				
	1846	1872	1930	1940	1950
	(1)	(2)	(3)	(4)	(5)
Cash Crop	2.20	1.53	0.08	0.00	0.16
Manufacturing	0.06	0.06	1.97	1.57	1.78
Processed Food	0.45	0.09	0.24	0.43	0.88
Raw Material	0.66	0.28	0.92	0.12	0.11
Staple	0.46	0.38	1.56	0.45	1.39

Figure 4: Structural change in São Paulo

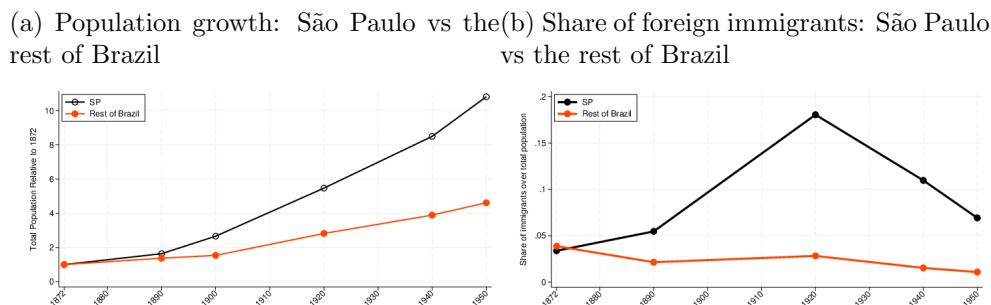


its agricultural share fell also much faster than the rest of the country.⁹ Appendix Figure 15 shows that between 1872 and 1950 São Paulo accumulated about a third of Brazil’s total employment in manufacturing.

4.1.4 São Paulo’s population grew and the region received a large inflow of international migrants

During this period, population grew substantially in Brazil, but more so in São Paulo. The census data shows that total population in São Paulo grew by a factor of 8 between 1872 and 1950, while it grew only by a factor of 5 in the rest of Brazil. This faster growth in São Paulo was fueled, especially until the 1920s, by an inflow of immigrants from Europe (see Figure 5). About 2.5 million immigrants moved to São Paulo during this period.

Figure 5: Population growth and Immigration



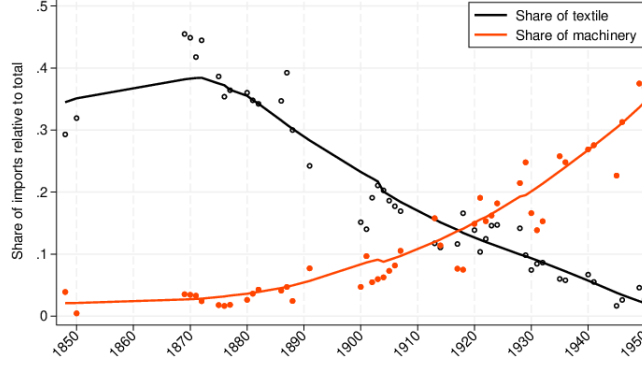
4.1.5 Brazil Substitutes Textile Imports for Own Production

The leading growth sectors in São Paulo’s manufacturing during this period were textile production and food processing. Figure 6 shows that the domestic manufacturing expansion coincided with a displacement of imports of textiles from the rest of the world, which drop to nearly zero, as Brazil starts importing heavy machinery and substitutes textile imports for its own production (Suzigan, 1986).¹⁰

⁹Appendix Figure 14 shows that between 1900 and 1920, São Paulo’s manufacturing output increased by a factor of 4.

¹⁰Textiles include linen, wool, and cotton goods, both processed and raw. Machinery covers locomotives, electrical equipment, and tools such as medical and scientific instruments. Combined, textiles and manufactures represent 40 to 50 percent of Brazil’s imports throughout the period. Food products are also a major import category, notably wheat derivatives, olive oil, and wine — none of which were produced in Brazil. Finally, another major import item is coal.

Figure 6: Textile Imports from Abroad and Local Production



Source: Own digitization of the Statistics of Commerce.

4.2 Micro Facts

We next explore the dual role that coffee suitability played in specialization and industrialization. Specifically, we estimate the following equation

$$y_{it} = \delta_i + \delta_t + \beta \times \text{Own Treat}_{it} + \delta \times \text{Mkt Access Treat}_{it} + \gamma X_{it} + \epsilon_{it}. \quad (1)$$

In this equation, i denotes a municipality, t denotes time, y_{it} denotes a set of outcomes, including sectoral employment shares, total and foreign population, and textile manufacturing outcomes. We create two treatment effects based on regional share of terra roxa. First, own treatment:

$$\text{Own Treat}_{it} = (\text{Share of Terra Roxa in } i's \text{ Land}) \times \ln \text{Coffee Exports}_t^{BRA}$$

interacts local terra roxa availability with export growth of Brazil as a whole. Second, market access treatment:

$$\text{Mkt Access Treat}_{it} = \left(\underbrace{\sum_{j \neq i} \tau_{ji,1872}^{-1} \cdot \text{Share of TR}_j}_{\text{Mkt Access}_i} \right) \times \ln \text{Coffee Exports}_t^{BRA},$$

which computes a measure of access to terra roxa in other regions, inversely weighted by distance according to the 1872 transportation network, i.e., before the coffee boom. Because we want to isolate the effect of foreign commodity demand on local industrial development,

we instrument Brazil’s exports with coffee production from the rest of the world.¹¹ Finally, for ease of interpretation, we express the variables “Share of Terra Roxa in i ’s Land” and “Mkt Access $_i$ ” as percentiles of their distribution before multiplying them by $\ln \text{Coffee Exports}_t^{BRA}$.

Sectoral employment shares. Table 2 shows the relation between coffee suitability and labor specialization across sectors. The population census is available for the years 1872 and 1920, a period during which Coffee Exports grew by 1.17 log points. The table shows a main pattern in Columns (2), (4), and (6): Structural change responds, separately, to local availability of terra roxa and to proximity to regions with terra roxa. Column (2), for example, shows that in a region in the 100th percentile of terra roxa availability, manufacturing employment shares grew by about 2 p.p. less than in a region in the 0th percentile. In contrast, in a region in the 100th percentile of the “market access” distribution, the manufacturing share grew by more than 5 p.p. more (compared to one at the bottom of the distribution). These are large relative changes, considering that in the initial period of 1872 the average manufacturing share across municipalities in Brazil was 5 percentage points. Column (6) shows comparable patterns for the service employment share, while Column (4) shows the converse for the agricultural employment. For both services and agriculture, the coefficients are also quantitatively large. As Columns (1), (3), and (5) show, identifying the effect of terra roxa availability requires separating that of own availability against that of proximity in other regions.

Migration and Population. Table 3 shows the impact of our two treatments on the local stock of migrants and total population. The sample includes the years 1872, 1890, and 1920, so we also include our IV specifications. As Columns (1)-(4) show, both own suitability and access to other regions with high suitability have a positive impact on the stock of foreign population. Focusing on our 2SLS estimates in Column (4), (i) a region in the 100th percentile of terra roxa availability, the stock of foreign population grew by about 0.6 log points more than in a region in the 0th percentile and (ii) compared to a region in the bottom of the distribution, a region in the 100th percentile of the “market access” distribution about 2.0 log points faster growth in its foreign-born population. Columns (5)-(8) show that the stock of total population responds only to local terra roxa availability, with a similar magnitude as the response of foreign population.

Textile Manufacturing Census Outcomes. We now turn to a newly digitized element in our data set—data from the textile manufacturing census, which are available for the years 1853,

¹¹We only do so when our regressions have three or more observations on the time dimension, since otherwise the instrument and the independent variable are perfectly correlated.

Table 2: Terra Roxa and Sectoral Employment Shares

	Manufacturing Shr.		Agriculture Shr.		Service Shr.	
	(1)	(2)	(3)	(4)	(5)	(6)
Own Treat	0.00454 (0.006)	-0.0185* (0.007)	0.00631 (0.012)	0.0570*** (0.016)	-0.0109 (0.007)	-0.0385*** (0.010)
MA Treat		0.0513*** (0.010)		-0.113*** (0.023)		0.0616*** (0.015)
Rsqr	0.755	0.768	0.850	0.857	0.914	0.917
N	1224	1224	1224	1224	1224	1224
IV	No	No	No	No	No	No
Region, Year FE.	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table presents the results of regression 1 with sectoral employment shares as dependent variables. The regression includes the years 1872 and 1920.

Table 3: Terra Roxa, Population, and Migration

	Ln Foreign				Ln Population			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Own Treat	1.931*** (0.238)	0.762** (0.268)	1.597*** (0.249)	0.664* (0.288)	0.496*** (0.071)	0.513*** (0.085)	0.458*** (0.071)	0.503*** (0.085)
MA Treat		2.692*** (0.286)		2.171*** (0.293)		-0.0405 (0.094)		-0.104 (0.095)
Rsqr	0.820	0.833	0.199	0.256	0.911	0.911	0.164	0.164
N	1630	1630	1630	1630	1630	1630	1630	1630
IV	No	No	Yes	Yes	No	No	Yes	Yes
Region, Year FE.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table presents the results of regression 1 with total population foreign population and as dependent variables. The regression includes the years 1872, 1890, and 1920.

Table 4: Terra Roxa and Textile Census Outcomes

	No. of Firms		No. Workers		No. of Looms	
	(1)	(2)	(3)	(4)	(5)	(6)
Own Treat	-0.0999*	-0.171	-78.32**	-96.49**	-57.49**	-73.13**
	(0.046)	(0.122)	(29.094)	(34.717)	(21.906)	(26.142)
MA Treat	0.261***	0.370	104.2**	136.3**	85.42**	113.0***
	(0.061)	(0.191)	(38.156)	(45.531)	(28.729)	(34.284)
Rsq	0.493	0.00606	0.395	0.00350	0.398	0.00371
N	3085	3085	3085	3085	3085	3085
IV	No	Yes	No	Yes	No	Yes
Region, year FE	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	No	No	No	No	No	No

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table presents the results of regression 1 with sectoral employment shares as dependent variables. The regression includes the years 1853, 1866, 1885, 1905, 1919.

1866, 1885, 1905, 1919. Table 4 shows results for three dependent variables: the number of textile firms, the number of workers employed in those firms, and the number of looms. Across the board, the table shows similar effects of own suitability and access to suitability in other regions. Focusing on Column (4), for example, which contains our 2SLS specification, we find that the increase in the number of textile workers was smaller by about than 130, when comparing a region at the top to a region at the bottom of the distribution of own availability; in contrast, the increase was larger by close to 200 workers, when looking at the “market access” distribution (note that over this period, $\Delta \ln \text{Coffee Exports}_t = 1.4$). These are quantitatively large results, considering that in 1919 the average number of workers across municipalities was 126. The table also shows that results are quantitatively large and in the same direction for the number of firms and the number of looms, although in our 2SLS the impact on the number of firms is measured imprecisely.

Manufacturing productivity. In our last exercise, we evaluate the relation between coffee suitability and manufacturing productivity. The manufacturing census is only available starting in 1940, so in our regressions we stack both 1940 and 1950 (so as to gain precision). Using the census, we construct a coarse measure of labor productivity, taking the ratio between revenues and total employment in each municipality. Because the coffee boom has largely already occurred by 1940, our regression does not exploit purely time-series variation, and we do not include region fixed effects. That is why our full specification, in Column

Table 5: Terra Roxa and Manufacturing Productivity

	ln Revenues / Worker		
	(1)	(2)	(3)
Own Treat	0.0320*** (0.006)	-0.0150* (0.008)	-0.0286*** (0.007)
MA Treat		0.107*** (0.010)	0.115*** (0.010)
Ln Population			0.153*** (0.032)
r2	0.485	0.547	0.570
N	1146	1146	1146
IV			
Standard errors in parentheses			
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$			

Notes: This table presents the results of regression 1 with revenue per worker as dependent variable. The regression includes the years 1940 and 1950.

(3), controls for regional characteristics that might be correlated with productivity by including log population as an additional control. In that column, we observe results that are consistent with our previous micro facts: (i) A one-standard-deviation increase in the own treatment variable leads to 0.03 log points lower labor productivity in manufacturing, while (ii) a one-standard-deviation increase in the market-access treatment leads to 0.12 log points larger labor productivity in manufacturing.¹²

Final remarks. In this section we have documented a collection of micro facts that demonstrate the dual role of coffee availability on industrial development. Regions with high local coffee suitability experienced slower growth in manufacturing employment (including in the leading textile sector) and by the end of the period have lower manufacturing productivity. Regions that are close to other regions with high coffee suitability, in contrast, experienced faster manufacturing employment growth and by the end of the period have substantially higher manufacturing productivity. In addition, both local suitability and proximity to high-suitability regions lead to large foreign immigration and population growth. We next turn to a standard, spatial economics model to understand what are the key features that a theory must have to rationalize these and our macro facts.

¹²In 1940, the standard deviations of the two treatments are 3.5 and 2.7.

5 Model

We now develop a spatial, open economy model with which we study the drivers of manufacturing development. The goal of the model is to assess (i) how important was the coffee boom in São Paulo’s industrialization and (ii) what are the key elements that such a model needs to fit both micro and macro facts.

5.1 Environment.

The economy is divided into home and foreign. Home is composed of I regions. There are four sectors in the economy, indexed by $k \in K$: manufacturing, M , food, F , coffee, C , and services S . Each region is endowed with Labor, N_i , and land, L_i , which can adjust endogenously due to migration and expansions in the land frontier. The local wage is w_i and the rental rate of land is r_i . All sectors except services are tradeable—both domestically and abroad—, subject to iceberg trade costs $d_{ij,k} \geq 1$.

Preferences. There is a representative consumer in region i , whose utility is captured by a three-tier CES system. The upper tier aggregates agriculture, manufacturing, and services, with constant elasticity σ . The middle tier aggregates coffee and food, with constant elasticity κ . The lower tier aggregates regionally differentiated varieties to produce an aggregate for each of the sectors, with constant elasticity η_k .

Technology. The marginal cost of production in all sectors k follows a Cobb-Douglas technology given by

$$c_{ik} = \frac{(w_i^{\gamma_k} r_i^{1-\gamma_k})^{1-\alpha_k} P_i^{\alpha_k}}{z_{ik}},$$

where γ_k is the labor share of value added, α_k is the share of intermediate inputs in production, P_i is the price of the intermediate input bundle, which coincides with the ideal price index of consumers, and z_{ik} is a productivity shifter.

Labor supply. There is a fixed population in the world, given by $\bar{N}$, which sorts across the world according to the following labor supply function¹³

$$N_i = \frac{\bar{u}_i (w_i/P_i)^\varsigma}{\sum_{i'} \bar{u}_{i'} (w_{i'}/P_{i'})^\varsigma} \bar{N}.$$

¹³This labor supply can be micro-founded by assuming workers have idiosyncratic, T1EV taste shocks across regions. The location parameter of the distribution of shocks is $\bar{u}_i$, while the inverse of the dispersion parameter is ς . Workers choose their location so as to maximize their real wages.

Land supply. Land supply in each region is given by the following function:

$$L_i = g_i^{\frac{\zeta}{\zeta-1}} \left(\frac{r_i}{\zeta P_i} \right)^{\frac{1}{\zeta-1}}$$

where g_i is a land productivity shifter.¹⁴

5.2 Equilibrium

Given our assumption on Armington differentiation, the price index of sector k in destination j is given by:

$$P_{j,k} = \left(\sum_i (d_{ij,k} c_{i,k})^{1-\eta_k} \right)^{\frac{1}{1-\eta_k}}.$$

We define the middle-tier price index, $\tilde{P}_{j,k}$, as follows:

$$\tilde{P}_{j,k} = \begin{cases} \left(\sum_{k \in \{C,F\}} \tilde{b}_{j,k} (P_{j,k})^{1-\phi} \right)^{\frac{1}{1-\phi}} & \text{if } k \in \{A\} \\ P_{j,k} & \text{if } k \in \{S, M\} \end{cases},$$

because Manufacturing and Services are not further disaggregated into subsectors. Finally, the aggregate price index in location j is given by

$$P_j = \left(\sum_{k \in \{S,M,A\}} b_{j,k} \left(\tilde{P}_{j,k} \right)^{1-\sigma} \right)^{\frac{1}{1-\sigma}}.$$

Note that these same price indexes apply to firms using goods as intermediate inputs.

Our formulation for preferences and technology also imply that in destination j expenditure shares for consumption and as intermediate inputs sourced from country i satisfy:

$$\pi_{ij,k} = \frac{b_{ij,k} (d_{ij,k} c_{i,k})^{1-\eta_k}}{P_{j,k}^{1-\eta_k}}.$$

Expenditure shares in broad sectors of the economy, in turn, satisfy

$$\beta_{i,k} = \frac{\tilde{b}_{i,k} \left(\tilde{P}_{i,k} \right)^{1-\sigma}}{P_i^{1-\sigma}} \text{ if } k \in \{S, A, M\},$$

¹⁴Pellegrina and Sotelo (2025) show how to obtain this formulation from a technology that combines an intermediate input bundle and a local resources owned, for example, by a government that develops land so as to maximize the profits from the sale of land.

and within broad sectors, we define

$$\tilde{\beta}_{i,k} = \begin{cases} \frac{\tilde{b}_{i,k} (P_{i,k})^{1-\phi}}{\tilde{P}_{i,k}^{1-\phi}} & \text{if } k \in \{C, F\} \\ 1 & \text{if } k \in \{S, M\} \end{cases}.$$

Note that the shares β_{ik} and $\tilde{\beta}_{ik}$ apply to both expenditures from households and as intermediate inputs in production.

Household total income in region i is

$$I_i = w_i N_i + r_i L_i,$$

which aggregates payments to factors of production.

Household expenditure in sector k is given by:

$$E_{i,k}^H = \beta_{i,k} \tilde{\beta}_{i,s(k)} I_i,$$

while expenditure coming from intermediate input use in sector k is given by:

$$E_{i,k}^F = \beta_{i,k} \tilde{\beta}_{i,s(k)} \sum_{k'} \alpha_{k'} X_{i,k'},$$

where $X_{i,k}$ is total sales of country i in sector k , given by

$$X_{i,k} = \sum_j \pi_{ij,k} E_{j,k},$$

where $E_{j,k} = E_{j,k}^H + E_{j,k}^F$.

Equilibrium definition. An equilibrium in this economy is a set of factor prices $\{w_i, r_i\}$ such that labor and land markets clear

$$\begin{aligned} w_i \frac{\bar{u}_i (w_i/P_i)^\zeta}{\sum_{i'} \bar{u}_{i'} (w_{i'}/P_{i'})^\zeta} \bar{N} &= \sum_{k \in \{S, C, F\}} (1 - \alpha_k) \gamma_k X_{i,k} \\ r_i g_i^{\frac{\zeta}{\zeta-1}} \left(\frac{r_i}{\zeta P_i} \right)^{\frac{1}{\zeta-1}} &= \sum_k (1 - \alpha_k) (1 - \gamma_k) X_{i,k} + (1 - \alpha_M) (1 - \gamma_M) (X_{i,M} - \Pi_{i,M}^{net}). \end{aligned}$$

5.3 Calibration

Our calibration approach combines standard model-inversion procedures with parameters we select from prior literature. Because it is the earliest period for which all the data we need are available, we calibrate our model to the year 1920, and in our counterfactual exercises we consider the impact of removing shocks.¹⁵

Elasticities and shares from the literature. From the literature, we pick $\sigma = 0.5$ (Comin, Lashkari, and Mestieri, 2021), $\phi = 3$ (Sotelo, 2020), and $\eta_k = \eta = 4$ (Simonovska and Waugh, 2014). We use as migration elasticity $\varsigma = 2$, which is the international elasticity in Fajgelbaum and Redding, 2022 and $\zeta = 1.5$, consistent with Pellegrina and Sotelo (2025). We then pick the share of land γ_k to be zero in non-agriculture and 0.8 in agriculture.

Matching observed data. We calibrate the intermediate input share, α_k , as follows. For agriculture, we use the agricultural census from 1920, which contains information on total costs of inputs (such as fertilizers and machinery) and revenues, to measure the cost share of agricultural inputs, giving a share of 0.07.¹⁶ For manufacturing, we calibrate the share of intermediates in the model so that it matches the share of workers in manufacturing in 1920 in Brazil in the data, which gives a share of intermediates of 0.77. Finally, for services, we pick 0.3, which is consistent with the World Input-Output Tables. Given the values for α_k and γ_k , we choose $z_{i,k}$ so as to match data on gross output per sector and region in the data. We pick values for $\bar{u}_i$ to match the number of workers in each region in equilibrium, and we pick $\bar{N}$ directly from data.

We adopt a hub-and-spoke parameterization of trade costs that allows us to calibrate bilateral trade cost values by targeting aggregate trade flows. In particular, we assume

$$d_{ij,k} = \delta^0 \left[(dist_{ij})^{\delta^1} \right]^{\iota_{ij}^D} \left[\delta_{kt} (dport_i)^{\delta^1} (dport_j)^{\delta^1} \right]^{1-\iota_{ij}^D},$$

where $dist_{ij}$ is the distance in hours between two regions i and j accounting for domestic infrastructure availability in 1920, $dport_i$ is the minimum distance to the closest port, ι_{ij}^D equals 1 if i and j belong to the same country and 0 otherwise; δ^0 is chosen to match total domestic trade relative to total internal trade $\sum_{rr} X_{rr}/X_{HH}$; δ^1 picked from the literature (0.05 in Pellegrina and Sotelo, 2025), and δ_{kt} is picked to match trade with ROW.

¹⁵Fajgelbaum and Redding (2022) and Pellegrina and Sotelo (2025) follow a similar approach when quantitatively assessing the role of international trade in historical episodes.

¹⁶This low cost share of intermediate inputs in agriculture is consistent with the cost share of inputs of low-income countries, see Farrokhi and Pellegrina (2023).

5.4 Untargeted moments

The model reproduces São Paulo’s RCA in manufacturing, which we do not target explicitly. In the data, the RCA in 1930 is equal to 1.98, while our simulation delivers a value of 1.4, meaning that the region’s regional export basket is disproportionately specialized in manufacturing, relative to the rest of Brazil.

6 Counterfactual analysis

We consider a sequence of counterfactual scenarios aimed at learning the extent to which the coffee shock by itself transformed São Paulo’s economy in a purely neoclassical economy, or whether other forces are necessary. In other words, we examine the strength of the Dutch Disease in our context and provide a justification for considering a Big Push type of mechanism in our next section.

All the scenarios we study below remove foreign demand for coffee, by reducing the value of the preference parameter $b_{ij,k}$ for $j = Foreign$ and $k = Coffee$, so that Brazil’s aggregate exports of coffee match those of 1872, i.e., at the outset of the coffee boom. From this basis, we consider four scenarios in which we manipulate the productivity of the manufacturing sector: (i) no change in productivity (“Neoclassical”), (ii) Marshallian externalities in the manufacturing sector (“IRS Labor”), (iii) Simultaneous manufacturing productivity reduction in São Paulo State (“Neoclassical w/ SP only”) and (iv) Simultaneous manufacturing productivity reduction as a function of access to red latosolo (“Neoclassical w/MA red latosol”).¹⁷

Results from these scenarios are shown in Figures 7 and 8, which compare how each counterfactual scenario accounts for our Macro and Micro facts. In each panel, the first bar shows the change in a variable that we observe between 1872 and 1920, while the next four show what each counterfactual scenario generates.

Beginning with Figure 7, we can see that a pure Dutch disease model has a difficult time rationalizing what we see: the boom in coffee demand generated a negligible increase in São Paulo’s manufacturing share, its RCA, and total population. For this calibration, growth in foreign demand increases the marginal product of land in coffee production, drawing it away from other sectors. At the same time, local income growth generates demand for nontraded

¹⁷Specifically, we parameterize manufacturing productivity as

$$z_{i,mfg,t} = \bar{z}_{i,mfg,0} \times e^{\varphi \times MA_{i,latosolo}},$$

and we pick φ so as to match aggregate manufacturing employment growth in Brazil, in the presence of the commodity boom.

services, precluding the manufacturing boom. The effects are quite similar when allowing for Marshallian externalities, as these make the demand for labor in manufacturing more elastic, inducing stronger quantity responses. By contrast, both models that simultaneously allow for localized growth in manufacturing productivity can account for a large fraction of manufacturing employment and RCA growth (less so with population growth).

Moving on to Figure 9, Panel (a) and (b) focus on the coefficients of “Own treatment” in our micro regressions, while Panels (c) and (d) focus on the coefficients of the “Market access treatment” variable. For this table, we replicate the same micro regressions under our alternative counterfactual scenarios and compare the outcomes to the data. The message here is similar in that it shows that absent simultaneous productivity growth in manufacturing is unable to replicate our micro facts. Panels (b), (c), and (d) show that the model with productivity growth correlated with market access can account for all correlations, except that of own latosoil and population growth. All other models, especially the Dutch disease ones, are unable to do so.

7 Big Push and Commodity Booms

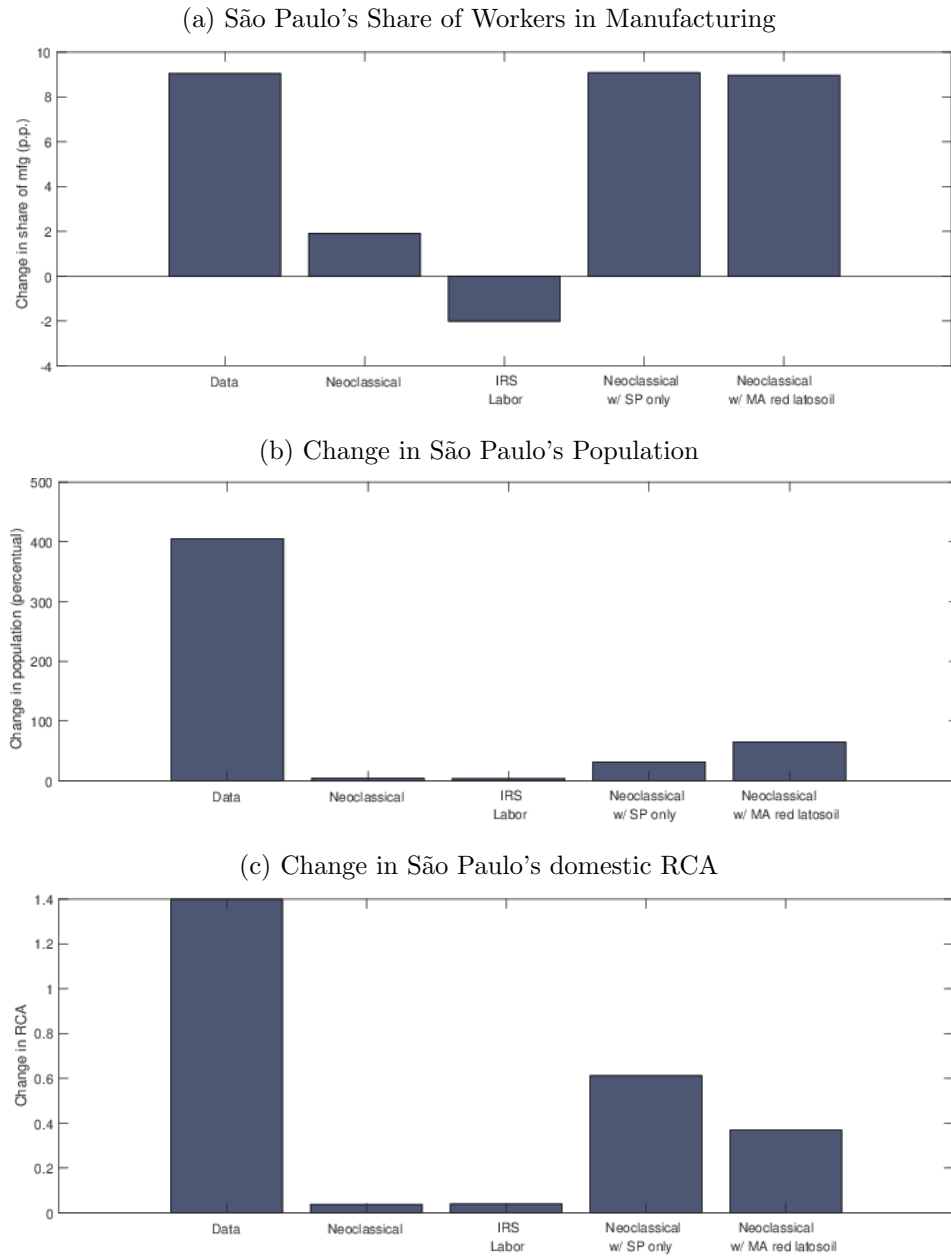
Motivated by the results in the previous section, we study a variant of the model that allows us to endogeneize the manufacturing sector’s productivity in response to the coffee boom. Appendix D explains how we modify our model to introduce this mechanism. Broadly, we introduce two main changes. The first is that we adopt the Big Push formulation by Murphy, Shleifer, and Vishny (1989b) for our manufacturing sector. The second is that we restrict the geography of our model to have only three regions: São Paulo State, Rest of Brazil, and Rest of the World. Moreover, we restrict attention to the possibility of multiple equilibria in the State of São Paulo.

Figure 9 shows, on the vertical axis, the individual incentives of firms, as a function of foreign coffee demand, both when all other firms industrialize (in blue) and when they do not (in black). The Figure shows that, absent the coffee boom (i.e., to the left) both equilibria are possible; however, as domestic expenditure grows due to the coffee income, all firms find it profitable to industrialize, regardless of what other firms do.

To understand the role of other fundamentals, we evaluate the impact of the coffee shock under different calibrations of our model. The results are shown in Figure 10. Panel (a) shows that if there is no migration, there is no coordination problem. This means that the commodity boom is incapable of explaining industrialization.¹⁸ Panel (b), in turn, highlights

¹⁸Recall the way we compute our counterfactuals is by starting in 1920, when the population in São Paulo is already large. Thus, no migration means that the population in São Paulo is large, and the demand for

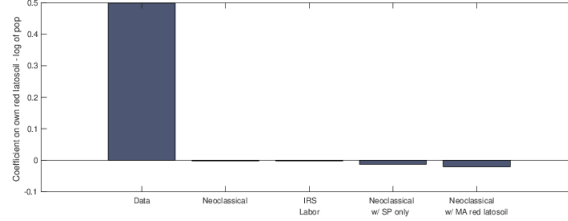
Figure 7: Macro-Economic Facts



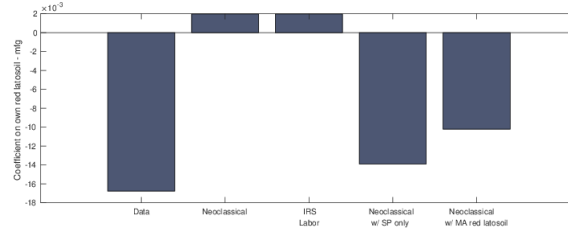
Notes: This figure compares the observed changes in selected macroeconomic variables in the state of São Paulo with the corresponding changes generated by different model specifications in which foreign demand for coffee is removed. Each panel focuses on a distinct macroeconomic outcome: panel (a) shows the change in the share of workers employed in manufacturing in São Paulo, panel (b) shows the change in São Paulo's total population, and panel (c) shows the change in São Paulo's domestic revealed comparative advantage (RCA).

Figure 8: Micro-Economic Facts

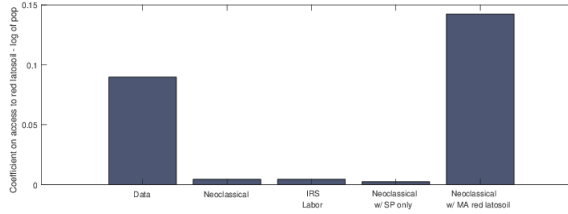
(a) Own red latosoil — Log of total pop



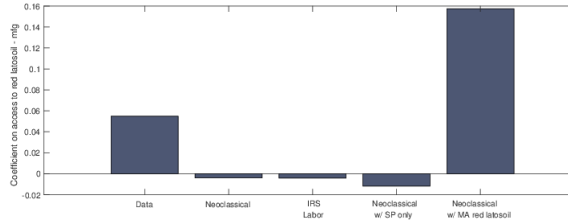
(b) Own red latosoil — Share of mfg workers



(c) Access to red latosoil — Log of total pop



(d) Access to red latosoil — Share of mfg workers



Notes: This figure compares the coefficients estimated in the data for changes in different dependent variables between 1872 and 1920 with the corresponding changes generated by the model simulations. Each panel focuses on a distinct dependent variable: panel (a) and (c) shows the change in the log of total population, and panel (b) and (d) shows the change in the share of workers employed in manufacturing. Panel (a) and (b) present the coefficient on own red latosoil and Panel (c) and (d) on access to red latosoil.

Figure 9: Commodity booms shifting across equilibria

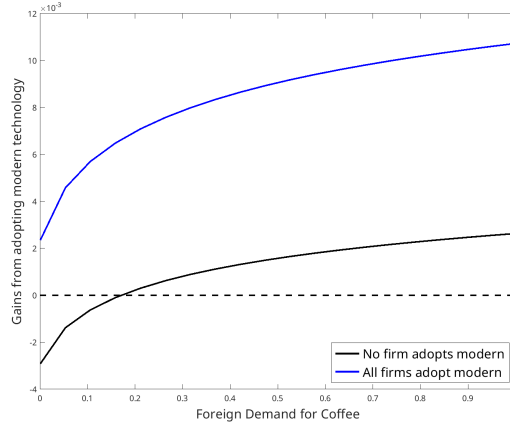
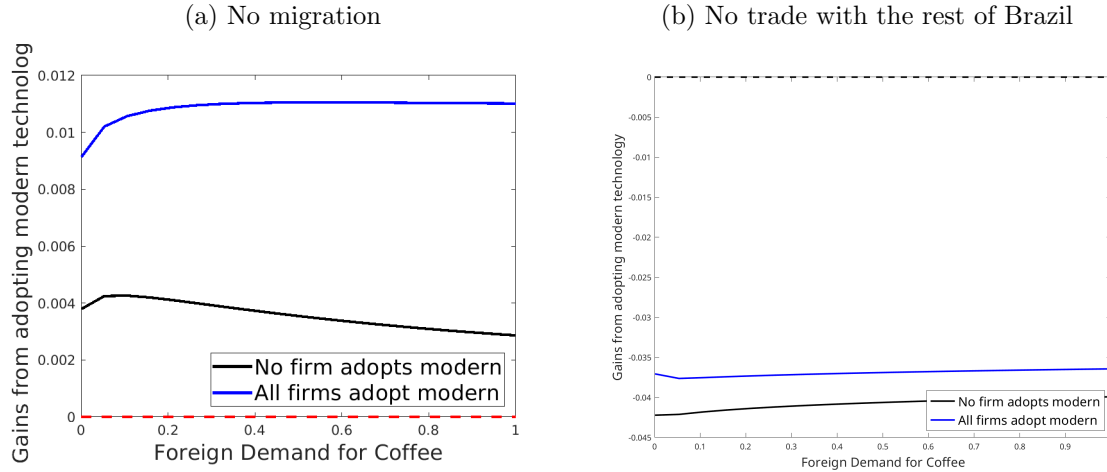


Figure 10: Commodity booms shifting across equilibria



the role of internal trade in allowing for São Paulo's industrialization. If domestic trade costs are recalibrated so as to eliminate manufacturing trade within Brazil, there is no industrialization in São Paulo, regardless of the commodity boom.

8 Conclusion

We reexamine the link between commodity booms and industrialization. We have empirically documented a dual role for the coffee boom in São Paulo's industrialization: it slowed down local industrialization in regions with high coffee suitability, but provided a demand base for industry located in surrounding areas. Our quantitative analysis shows that in the presence of manufacturing is large, regardless of the coffee boom.

Big Push dynamics, coffee growth can shift the economy to an industrialization equilibrium. Key to this conclusion are the possibility of migration from abroad, which relaxes factor endowment constraints, and relatively low domestic trade costs, which allow industry to expand domestically.

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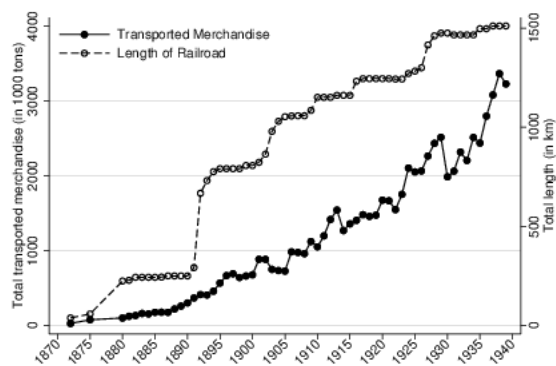
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A Additional Figures

Figure 11: Railroad Expansion in São Paulo

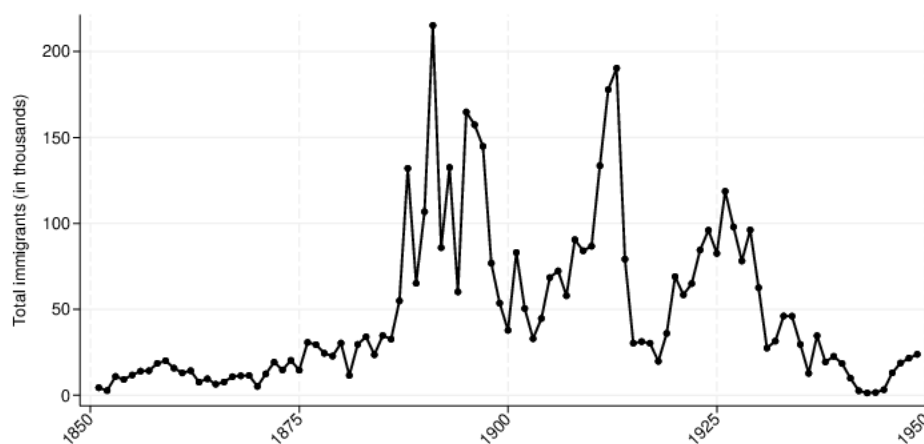


Notes: Digitized data come from Cia Paulista de estradas de Ferro.

Figure 12: Example of “Terra Roxa”

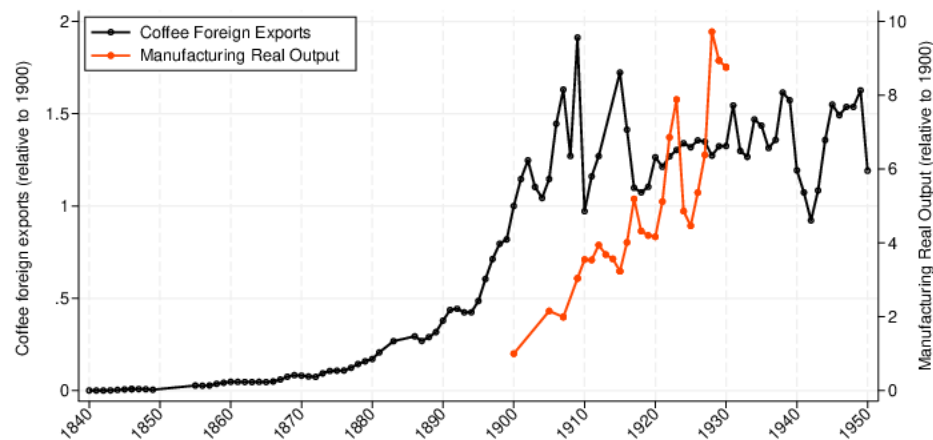


Figure 13: Total Immigration into Brazil



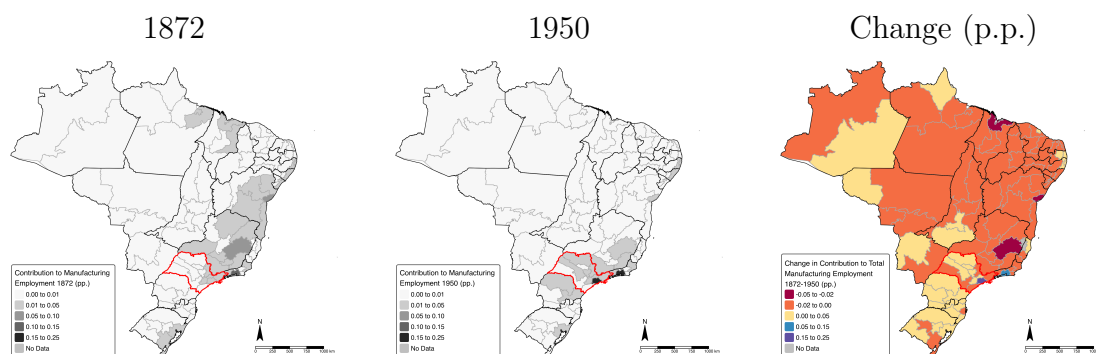
B Appendix to Macro Facts

Figure 14: São Paulo: Trade and Manufacturing Real Output



Note: Total production before 1900 is unavailable

Figure 15: São Paulo concentrates manufacturing



C Appendix to Micro Facts

Figure 16: A peak at the variation

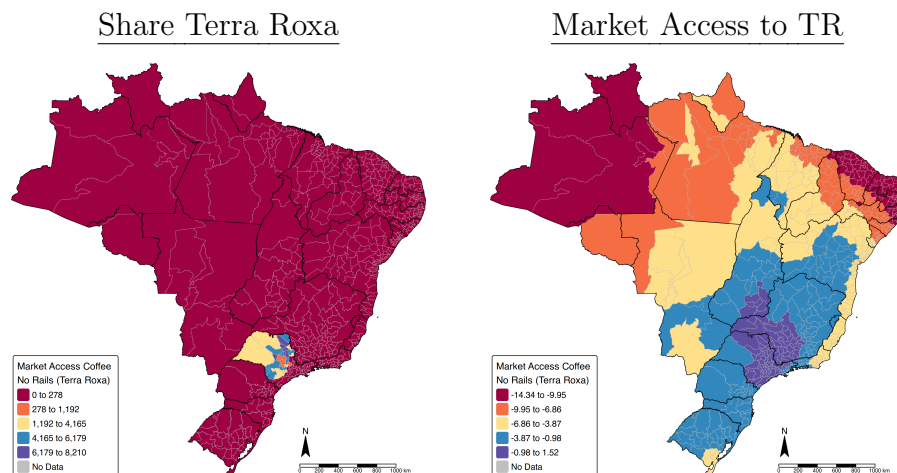


Table 6: Terra Roxa and Coffee Production

	Ln Production of Coffee		Ln Coffee Area	
	(1)	(2)	(3)	(4)
Share red	2.942*** (0.341)	1.014* (0.405)	2.788*** (0.347)	0.892* (0.414)
MA		4.764*** (0.557)		4.694*** (0.599)
Rsqr	0.207	0.307	0.158	0.253
N	443	443	453	453
IV	No	No	No	No
Region, Year FE.	No	No	No	No
Covariates	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table presents the results of regressing coffee production on the own share of terra roxa and the market access of each region. Data are from the 1920 agricultural census.

Table 7: Terra Roxa and Coffee Area

	(1)	(2)	(3)	(4)
Own Treat	1.919*	5.531***	1.919*	5.531***
	(0.754)	(0.891)	(0.754)	(0.891)
MA Treat		-10.17***		-10.17***
		(1.539)		(1.538)
Rsq	0.943	0.951	0.0499	0.188
N	662	662	662	662
IV	No	No	Yes	Yes
Region, Year FE.	Yes	Yes	Yes	Yes
Covariates	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table presents the results of regressing coffee area on the own share of terra roxa and the market access of each region. Data are from the 1920 agricultural census.

Table 8: Terra Roxa and Long Differences in Sectoral Employment

	Manufacturing Shr.		Agriculture Shr.		Service Shr.	
	(1)	(2)	(3)	(4)	(5)	(6)
Own Treat	0.00687	-0.0344***	0.00160	0.0859***	-0.00847	-0.0515***
	(0.008)	(0.010)	(0.015)	(0.019)	(0.009)	(0.012)
MA Treat		0.0921***		-0.188***		0.0959***
		(0.013)		(0.027)		(0.018)
Rsq	0.659	0.696	0.759	0.782	0.837	0.847
N	1224	1224	1224	1224	1224	1224
IV	No	No	No	No	No	No
Region, Year FE.	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table presents the results of regression 1 with sectoral employment shares as dependent variables. The regression includes the years 1872 and 1950. Over this period, $\Delta \ln \text{Coffee}_t = 1.17$.

Table 11: Terra Roxa and Intensive Textile Census Outcomes

	(1)	(2)	(3)	(4)
	ln Looms / Worker	ln Spindles / Worker	ln HP / Worker	ln Capital / Worker
Own Treat	0.144	-0.330	-0.375	-0.858
	(0.331)	(0.538)	(1.315)	(0.472)
MA Treat	0.806	2.033*	1.897	2.803**
	(0.561)	(0.790)	(0.987)	(0.814)
r2	0.615	0.633	0.640	0.705
N	123	111	99	114
IV	No	No	No	No
FE	Yes	Yes	Yes	Yes
BC	No	No	No	No

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table presents the results of regression 1 with sectoral employment shares as dependent variables. The regression includes the years 1853, 1866, 1855, 1905, 1919.

Table 9: Terra Roxa and Long Differences in Population and Migration

	Ln Population				Ln Foreign			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Own Treat	0.458*** (0.074)	0.566*** (0.093)	0.458*** (0.074)	0.566*** (0.093)	1.689*** (0.182)	0.775*** (0.216)	1.689*** (0.182)	0.775*** (0.216)
MA Treat		-0.240* (0.100)		-0.240* (0.100)		2.110*** (0.268)		2.110*** (0.267)
Rsqr	0.887	0.888	0.161	0.169	0.830	0.849	0.289	0.368
N	1224	1224	1224	1224	1062	1062	1062	1062
IV	No	No	Yes	Yes	No	No	Yes	Yes
Region, Year FE.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table presents the results of regression 1 with total population foreign population and as dependent variables. The regression includes the years 1872 and 1950. Over this period, $\Delta \ln \text{Coffee}_t = 1.17$.

Table 10: Terra Roxa and Long Differences in Textile Census Outcomes

	No. of Firms		No. Workers		No. of Looms	
	(1)	(2)	(3)	(4)	(5)	(6)
Own Treat	-0.293** (0.112)	-0.481 (0.376)	-161.9* (64.848)	-235.7** (75.821)	-91.81** (28.964)	-127.0*** (33.868)
MA Treat	0.612*** (0.147)	0.941 (0.616)	245.3** (85.046)	370.8*** (99.436)	145.2*** (37.986)	206.6*** (44.417)
Rsqr	0.324	0.00394	0.323	0.00230	0.415	0.00431
N	3702	3702	3702	3702	3702	3702
IV	No	Yes	No	Yes	No	Yes
Region, year FE	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	No	No	No	No	No	No

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table presents the results of regression 1 with total population foreign population and as dependent variables. The regression includes the years 1853 and 1945. Over this period, $\Delta \ln \text{Coffee}_t = 1.7$.

Table 12: Terra Roxa and Long Differences in Intensive Textile Census Outcomes

	(1)	(2)	(3)	(4)
	ln Looms / Worker	ln Spindles / Worker	ln HP / Worker	ln Capital / Worker
Own Treat	-0.0353	-0.428	-0.375	-0.858
	(0.332)	(0.508)	(1.315)	(0.472)
MA Treat	0.813	1.720*	1.897	2.803**
	(0.552)	(0.726)	(0.987)	(0.814)
r2	0.530	0.521	0.640	0.705
N	207	199	99	114
IV	No	No	No	No
FE	Yes	Yes	Yes	Yes
BC	No	No	No	No

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table presents the results of regression 1 with sectoral employment shares as dependent variables. The regression includes the years 1853 and 1919.

D The Big Push Model

Changes relative to our quantitative model. There is a fixed mass of manufacturing firms in each region, M_i . Manufacturing firms have a choice of operating under a traditional technology ($m = 0$) and a modern technology, ($m = 1$). The traditional technology features constant returns to scale, as above. The modern technology requires paying a fixed adoption cost in units of labor, κ , but boosts productivity by a factor β_M . Inducing workers to work in the modern sector, however, requires paying them a premium ν (as in Murphy, Shleifer, and Vishny, 1989b). Pricing in manufacturing is monopolistically competitive, but firms in other sectors are perfectly competitive:

$$p_{i,k} = \begin{cases} \frac{\eta_k}{\eta_k - 1} \frac{c_i}{z_{i,M}} & \text{if } k = M \text{ and } m = 0 \\ \frac{\eta_k}{\eta_k - 1} \frac{c_i}{z_{i,M}} \frac{\nu_i^{\gamma_k}}{\beta_i} & \text{if } k = M \text{ and } m = 1 \\ \frac{c_i}{z_{i,k}} & \text{if } k \in \{S, C, F\} \end{cases}$$

Manufacturing firm decisions. Manufacturing firms choose to operate the new technology if their profits from doing so are larger than those of operating the old technology. Because all firms are symmetric, they will all make the same technology choice in a given region.